

Microsoft Foundry gpt-5.6-sol Configuration Guide

Configuration and operational guidance for the exact model deployment used by BidSight for quotation extraction and recommendation generation.

Document purpose

This guide documents the provider boundary implemented in `backend/app/services/foundry_service.py`, including the gpt-5.6-sol deployment, endpoint selection, structured outputs, error handling, security, and deployment checks.

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BidSight is a procurement decision-support system. Final purchasing authority remains with the authorised user.

1. Foundry's Role in BidSight

Microsoft Foundry has two responsibilities in the application:

Responsibility	Input	Output
Quotation extraction	Page-labelled text extracted from one PDF	Strict quotation fields, specifications, source pages, and missing information
Recommendation generation	Reviewed quotations plus Python-generated compliance, scores, ranks, risks, and missing information	Recommended vendor, concise reasoning, strengths, risks, missing information, and trade-off explanation

Non-negotiable boundary

Foundry does not calculate compliance, component scores, price scores, weighted totals, eligibility, or rank. Those values are produced and enforced by the Python scoring service.

2. Required Foundry Resources

- A Microsoft Foundry project and an accessible model deployment.
- A resource inference endpoint compatible with the OpenAI v1 API.
- An API key for that resource.
- The exact deployment name configured in Microsoft Foundry.

The current repository and environment example use gpt-5.6-sol as the deployment name.

FOUNDRY_MODEL_DEPLOYMENT must match the deployment name exactly; it is not inferred from the underlying model family.

Model capability requirement

The deployment must support structured chat-completion parsing for the Pydantic schemas used by the service. A model or endpoint that does not support the requested structured response will produce a controlled processing error.

3. Backend Environment Variables

```
FOUNDRY_ENDPOINT=https://YOUR-RESOURCE.openai.azure.com/openai/v1
FOUNDRY_API_KEY=YOUR_SECRET_KEY
FOUNDRY_MODEL_DEPLOYMENT=gpt-5.6-sol
FOUNDRY_REQUEST_TIMEOUT_SECONDS=90
```

Variable	Required	Purpose
FOUNDRY_ENDPOINT	Yes	Resource-level HTTPS endpoint used as the OpenAI SDK base URL.
FOUNDRY_API_KEY	Yes	Backend-only authentication secret.
FOUNDRY_MODEL_DEPLOYMENT	Yes	Exact deployment name passed as the model parameter.
FOUNDRY_REQUEST_TIMEOUT_SECONDS	No	Request timeout; defaults to 90 and is constrained to 1-300 seconds.

4. Use the Correct Endpoint

Accepted endpoint forms include:

```
https://YOUR-RESOURCE.services.ai.azure.com
https://YOUR-RESOURCE.services.ai.azure.com/openai/v1
https://YOUR-RESOURCE.openai.azure.com
https://YOUR-RESOURCE.openai.azure.com/openai/v1
```

When no path is supplied, the service appends /openai/v1/. When a path is supplied, it must end in /openai/v1.

Query parameters, fragments, embedded credentials, non-HTTPS URLs, project endpoints, and unrelated paths are rejected.

Resource endpoint versus project endpoint

Endpoint	Use in BidSight
Resource inference endpoint	Use this. It is compatible with the OpenAI SDK base_url configured by the backend.
Foundry project endpoint	Do not use it in FOUNDRY_ENDPOINT. The current service is not implemented with the Foundry project or agent SDK.
Playground or browser URL	Do not use it. It is a management page, not an inference base URL.

5. How the Client Is Created

```
OpenAI(  
    base_url=normalised_foundry_endpoint,  
    api_key=FOUNDRY_API_KEY,  
    timeout=FOUNDRY_REQUEST_TIMEOUT_SECONDS,  
)
```

Provider-specific logic is isolated in foundry_service.py. Routers call extract_quotation or generate_recommendation and translate service errors into controlled HTTP responses. No frontend component calls Foundry directly.

6. Quotation Extraction Contract

The extraction call uses client.beta.chat.completions.parse with FoundryQuotationExtractionResponse as the response format. The system instruction treats quotation text as untrusted document content and requires facts to remain faithful to the source.

Required extraction behaviour

- Never invent, estimate, infer, or assume a value or commercial term.
- Use null when a field is absent or unclear.
- List absent and ambiguous information explicitly.
- Return numbers without symbols or separators.
- Keep payment and support language concise and faithful.
- Capture technical attributes as name-value specification items.
- Return source page references where the document supports them.

Structured fields

vendor_name, product_name, product_model, quantity, unit_price, subtotal, total_price, tax, currency, delivery_days, warranty_months, payment_terms, support_details, quotation_validity_days, specifications, source_pages, and missing_information.

Persistence rule

The provider response is validated by Pydantic before it is converted into the application model and saved. A malformed response does not overwrite the quotation; the PDF remains available for retry.

7. Recommendation Contract

The recommendation call receives JSON evidence assembled by the evaluation router after deterministic scoring. It includes the evaluation baseline, requirements, reviewed quotation values, requirement outcomes, component scores, overall scores, ranks, statuses, failures, risks, and missing information.

Required recommendation behaviour

- Use only the supplied structured evidence.
- Do not recalculate or alter Python-generated values.
- Recommend only an eligible vendor present in the evaluation.
- Explain mandatory failures, uncertainty, risk, and cheaper-vendor trade-offs.
- Do not invent evidence or approve a purchase automatically.

Structured recommendation fields

Field	Meaning
recommended_vendor	Vendor name selected from eligible evidence.
concise_reasoning	Decision-focused summary.
strengths	Evidence-backed reasons supporting the selection.
risks	Relevant limitations or commercial concerns.
missing_information	Unresolved evidence that remains explicit.
tradeoff_explanation	Why a cheaper alternative was not selected when applicable.

After validation, the API independently verifies that the recommended vendor exists in the comparison and has COMPLIANT or PARTIALLY_COMPLIANT status. An invalid recommendation is rejected with HTTP 502.

8. Error Mapping and Recovery

Condition	Service error	API result	Recovery
Endpoint, key, or deployment missing	FoundryConfigurationError	503	Correct backend environment and redeploy.
Network, authentication, quota, timeout, or provider failure	FoundryRequestError	502	Review provider status and credentials; retry.
Empty, refused, or malformed structured response	FoundryResponseError	502	Keep the PDF, verify model capability, and retry.
Provider selects an absent or ineligible vendor	Router validation failure	502	Inspect evidence and model output; regenerate after correction.

9. Security Requirements

- Store FOUNDRY_API_KEY only in an untracked backend .env or encrypted cloud secret.
- Never prefix provider secrets with NEXT_PUBLIC_.
- Do not paste API keys into screenshots, documentation, source code, or client-side settings.
- Rotate a key immediately after accidental exposure.

- Use separate credentials or resources for development and production where possible.
- Limit access through the Foundry resource and hosting-platform permissions.
- Treat PDF text and structured evidence as untrusted input; the current prompts explicitly do so.

10. FastAPI Cloud Configuration

In the FastAPI Cloud app, set the application directory to backend and add:

```
DATABASE_URL=sqlite:///./bidsight.db
FOUNDRY_ENDPOINT=https://YOUR-RESOURCE.openai.azure.com/openai/v1
FOUNDRY_API_KEY=YOUR_SECRET_KEY
FOUNDRY_MODEL_DEPLOYMENT=gpt-5.6-sol
FOUNDRY_REQUEST_TIMEOUT_SECONDS=90
UPLOAD_DIR=uploads
FRONTEND_URL=https://YOUR-FRONTEND.vercel.app
MAX_UPLOAD_SIZE_MB=10
```

- Mark FOUNDRY_API_KEY as Secret before saving.
- Save and redeploy after any environment change.
- Set FRONTEND_URL to the exact Vercel production origin to satisfy CORS.
- In Vercel, set NEXT_PUBLIC_API_URL to the FastAPI Cloud application origin and redeploy.

11. Configuration Validation

Check	Expected result
GET /api/health	HTTP 200; this confirms the API is running, not that Foundry is callable.
Upload a sample quotation	Quotation record is created with UPLOADED status.
Process quotation	Status becomes READY and structured extraction is available.
Review and score all quotations	Python comparison is returned without a Foundry scoring call.
Generate recommendation	Structured recommendation names an eligible vendor from the comparison.
Inspect logs	No secret values or raw internal traceback are returned to the client.

12. Troubleshooting Reference

Message or symptom	Resolution
Use the Foundry resource endpoint, optionally ending in /openai/v1/.	Replace the project or management endpoint with the resource inference endpoint.
Microsoft Foundry API key is not configured.	Add FOUNDRY_API_KEY to the backend environment as a secret and redeploy.
Microsoft Foundry model deployment name is not configured.	Set FOUNDRY_MODEL_DEPLOYMENT to the exact deployment name.
Microsoft Foundry could not process the quotation.	Verify endpoint, key, deployment, model capability, quota, network access, and timeout.
Microsoft Foundry returned malformed structured data.	Use a deployment that supports the requested structured output and retry the retained PDF.
CORS failure in the browser	Set backend FRONTEND_URL to the exact frontend origin; redeploy both services after environment changes.

13. Final Operational Checklist

- Resource inference endpoint uses HTTPS and the supported path.
- API key is current, encrypted, and not exposed publicly.
- Deployment name matches Microsoft Foundry exactly.
- Configured model supports the structured response used by the application.
- All quotations are reviewed before evaluation.
- Recommendation is generated only after Python scores are stored.
- Frontend and backend origins are mutually configured.
- A live model call is tested with sample data before a demonstration.